

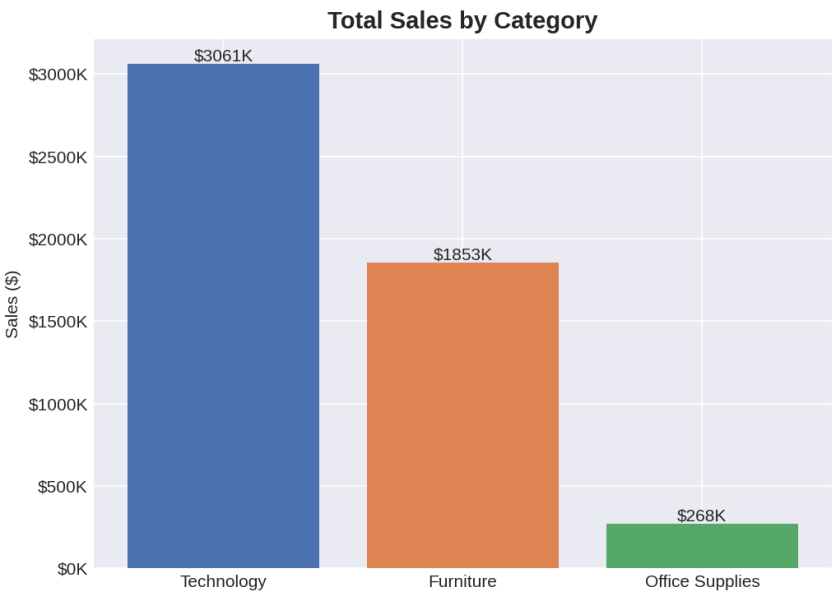
Data Analytics Internship — Final Report

Executive Summary | Superstore Sales Analytics Project

Executive Summary

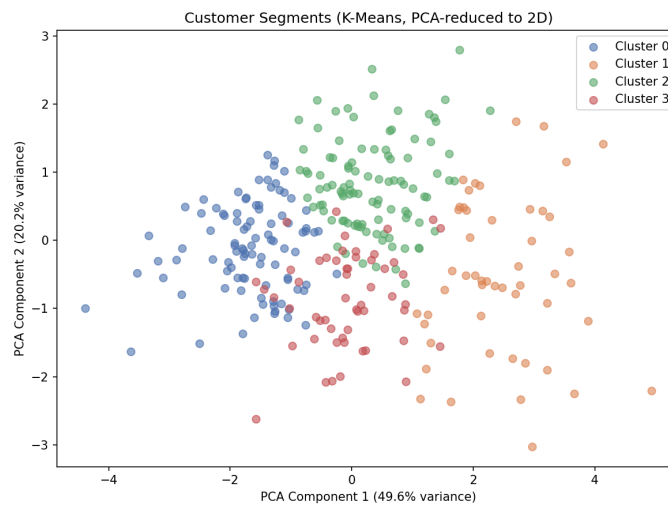
Over four tasks, this project delivered an end-to-end analytics workflow on a 2,500-record retail sales dataset: exploratory data analysis and SQL-based extraction, Python and BI-style dashboards, and advanced statistical modeling. The analysis surfaced clear revenue drivers, identified four distinct customer segments, and produced a working predictive model for sales — providing a foundation for data-driven decisions across inventory, marketing, and customer retention.

Key Visual — Sales by Category



Top 5 Insights

1. **Technology drives revenue.** Technology orders average \$3,723 in sales vs. \$312 for Office Supplies — a statistically significant gap ($p < 0.001$).
2. **Sales are right-skewed.** A small share of large orders account for a disproportionate share of revenue (skewness = 1.78).
3. **Four distinct customer segments exist.** K-Means clustering identified High-Value Champions (51 customers, \$30,965 avg. sales) down to an At-Risk segment (94 customers, \$8,002 avg. sales, highest discount reliance).
4. **Category preference is not segment-driven.** A chi-square test found no significant link between customer segment and category choice ($p = 0.996$) — other factors explain purchasing better.
5. **Sales are predictable to a useful degree.** A linear regression model explains 58% of sales variance ($R^2=0.582$, $MAE=\$1,158$), with order quantity, profit, and Technology category as the top three drivers.



Business Recommendations

1. **Prioritize Technology in inventory and marketing spend.** It is the highest per-order revenue driver and should anchor promotional and stocking decisions.
2. **Launch a targeted win-back campaign for the At-Risk customer segment.** Their high discount dependency paired with low overall spend signals churn risk — early intervention (loyalty offers, personalized outreach) can protect revenue.
3. **Use order-quantity-based bundling to lift sales.** Since quantity is the strongest predictor of sales, bundle offers or volume incentives are likely to have a measurable revenue impact.